EXPERIMENT 10: A* (A-STAR) ALGORITHM

AIM

To develop a Python program that implements the A* (A-Star) algorithm to find the shortest path between a start node and a goal node using heuristic search.

OBJECTIVE

- To understand the concept of the **A*** search algorithm in Artificial Intelligence.
- To find the shortest path from the start node to the goal node.
- To use heuristic values to improve search efficiency.
- To determine the optimal path with minimum cost.

ALGORITHM

Step 1: Start.

Step 2: Initialize the **Open List** with the start node and an empty **Closed List**.

Step 3: Select the node with the lowest **f(n) = g(n) + h(n)** value from the Open List.

Step 4: If the selected node is the goal node, display the shortest path and stop.

Step 5: Move the selected node from the Open List to the Closed List.

Step 6: Generate all neighboring nodes of the current node.

Step 7: Calculate the cost values:

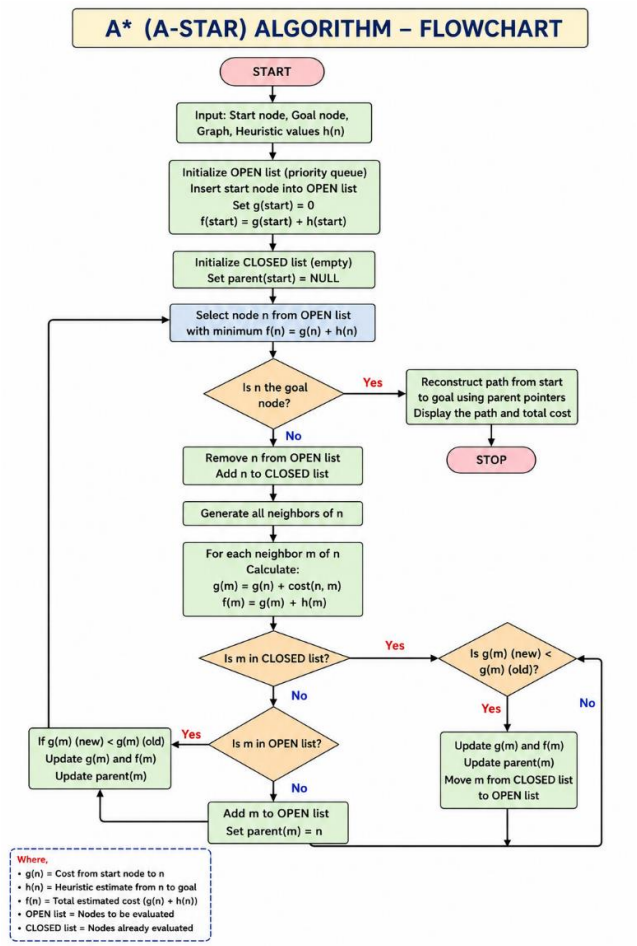
- **g(n)** = Cost from the start node.
- **h(n)** = Heuristic estimate to the goal.
- **f(n) = g(n) + h(n)**.

Step 8: Update the Open List with the neighboring nodes having the lowest cost.

Step 9: Repeat Steps 3–8 until the goal node is found or the Open List becomes empty.

Step 10: Stop.

FLOW CHART



SOURCE CODE

```
import heapq

graph = {
    'A': [('B', 1), ('C', 3)],
    'B': [('D', 3), ('E', 6)],
    'C': [('F', 5)],
    'D': [('G', 4)],
    'E': [('G', 2)],
    'F': [('G', 1)],
```

```
    'G': []
}

heuristic = {
    'A': 7,
    'B': 6,
    'C': 4,
    'D': 4,
    'E': 2,
    'F': 1,
    'G': 0
}

def astar(start, goal):
    open_list = []
    heapq.heappush(open_list, (0, start))

    g_cost = {start: 0}
    parent = {start: None}

    while open_list:
        _, current = heapq.heappop(open_list)

        if current == goal:
            path = []
            while current:
                path.append(current)
```

```
            current = parent[current]
        return path[::-1]

    for neighbor, cost in graph[current]:
        new_cost = g_cost[current] + cost

        if neighbor not in g_cost or new_cost < g_cost[neighbor]:
            g_cost[neighbor] = new_cost
            f_cost = new_cost + heuristic[neighbor]
            heapq.heappush(open_list, (f_cost, neighbor))
            parent[neighbor] = current

  return None

path = astar('A', 'G')

print("Shortest Path:", path)
```

OUTPUT

```
Shortest Path: ['A', 'B', 'D', 'G']
```

RESULT

The Python program successfully implements the A* (A-Star) algorithm and finds the shortest path from the start node to the goal node using heuristic values, producing the optimal path efficiently.